Task 1

What do u understand by Big O notation

- It is used to describe the time, and space complexity increases as the input size increases

Task 2

What are array ? how many types of Arrays ?

- An array is a data structure used to store multiple elements of the same data type in a single variable.
  - 1D Array
  - 2D Array
  - Muilt-dimensional array

Task 3

What do you understand about strings in java?

- A String is an object that represents a sequence of characters. It is one of the most commonly used classes and strings are Immutable.

Task 4

Create an array in java

```
public class Main {
   public static void main(String[] args) {
            int[] arr = new int[];
             arr[0] = 100;
              arr[1] = 200;
              arr[2] = 300;

       for (int i = 0 ;  i < arr.length ;  i++) {

           System.out.println(arr[i]);
       }
   }
}
```

Task 5

Create a java program to print the String in reverse order.

```
public class Main {
   public static void main(String[] args){

       String str = "MONIKA";
```

```
        System.out.println(" original string" + str);
        System.out.print("rev str : ");

        for(int i = str.length() - 1 ; i >= 0; i--){
          System.out.println(str.charAt(i));
        }
    }
}
```

Task 6

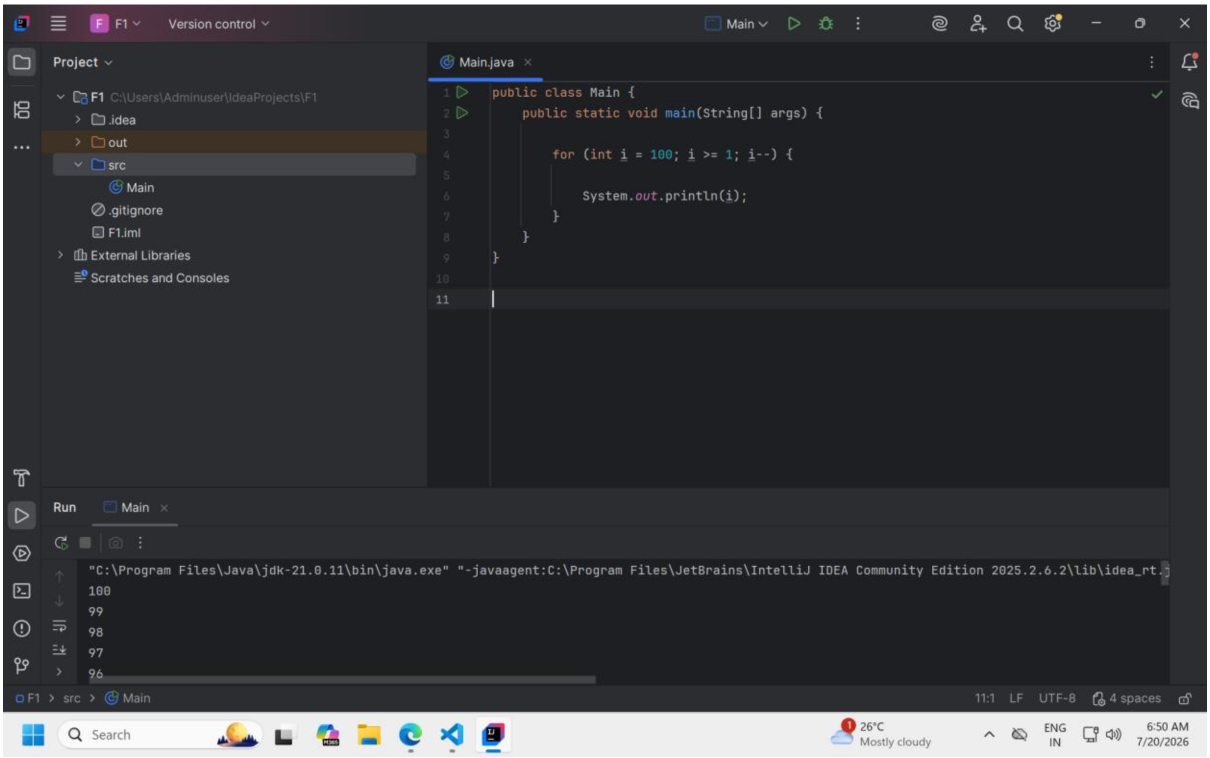
Write a program to declare static and normal variable and increment the values of both. What do you observe.

- Static variables are shared by all objects of a class, whereas normal or instance variables are unique to each object. When incremented, the static variable retains and shares its updated value across all objects, while the normal variable increments only for the specific object that accesses it.

```
Public class static_Normal{
        static int static_num = 0;
        int normal_num = 0;

        void increment(){
                static_num++;
                normal_num++;

                System.out.println("Static count:" + static_num + " | Normal Count: " +normal_num);
        }

        public static void main(String[] args){
              static_Normal obj1 = new static_num();
              static_Normal obj2 = new static_num();

              System.out.println("Calling from obj1 : ");
              obj1.increment();

              System.out.println("Calling from obj2 : ");
              obj2.increment();

              System.out.println("Calling obj1 again : ");
              obj1.increment();
              }
        }
```

Task 7

Write a program to print numbers from 100 to 10




```
public class Main {
    public static void main(String[] args) {

        for (int i = 100; i >= 1; i--) {

            System.out.println(i);
        }
    }
}
```

Task 8:

Write a program to accept login id and pwd from the user and display welcome message if Login id is “MONIKA” pwd is “43215”

USE While loop

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String loginId, password;

        System.out.print("Enter Login ID: ");
```

```
        loginId = sc.nextLine();

        System.out.print("Enter Password: ");
        password = sc.nextLine();

        while (!(loginId.equals("MONIKA") && password.equals("43215"))) {
            System.out.println("Invalid Login ID or Password!");

            System.out.print("Enter Login ID: ");
            loginId = sc.nextLine();

            System.out.print("Enter Password: ");
            password = sc.nextLine();
        }

        System.out.println("Welcome MONIKA !");
        System.out.println("Login Successful.");

        sc.close();
    }
}
```

Task 9 :

Write a program to accept login id and pwd from the user and display welcome message and then check for the login id and pwd ..

Use do while loop..

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String loginId, password;

        do {
            System.out.print("Enter Login ID: ");
            loginId = sc.nextLine();

            System.out.print("Enter Password: ");
            password = sc.nextLine();

            System.out.println("Welcome " + loginId);

        } while (!(loginId.equals("monika1733@gmail.com") && password.equals("12345")));

        System.out.println("Login Successful!");

        sc.close();
    }
}
```

Task 10:

Write a program to display the use of

If

```
public class Main {
    public static void main(String[] args) {

        int age = 20;

        if (age >= 18) {
            System.out.println("Eligible to Vote");


        }
    }
}
```

Else if

```
public class Main {
    public static void main(String[] args) {

        int age = 20;

        if (age >= 18) {
            System.out.println("Eligible to Vote");
            }
         else{
               System.out.println(“Not eligible”);


        }
    }
}
```

Nested if

```
public class Main {
    public static void main(String[] args) {

        int marks = 85;

        if (marks >= 35) {
            System.out.println("You Passed");

             if(marks >= 90){
                   System.out.println(“Grade: A+”) ;
             }
              else if(marks >= 75){
                      System.out.println(“Grade: A”);
               }
               else if(marks >= 60){
                      System.out.println(“Grade: B”);
                 }
                else{
```

```
                    System.out.println("Grade: C");
                }
            }
            else{
                System.out.println("Failed");
        }
    }
}
```

FOR EACH

```
public class ForeachLoop {
    public static void main(String[] args) {
        int arr[] = {1,3,5};
        for (int i : arr){
            System.out.println(i);
        }
        for (int i = 0; i <10; i ++) {
            System.out.println(i);
        }
    }

}
```

Task 11:

Write a program to display the use of switch case

```
import java.util.Scanner;

public class switch_case {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number (1-3): ");
        int choice = sc.nextInt();

        switch (choice) {
            case 1:

                System.out.println("You selected One");
                break;

            case 2:
                System.out.println("You selected Two");
                break;

            case 3:
                System.out.println("You selected Three");
```

```
            break;

        default:
            System.out.println("Invalid Choice");
        }

        sc.close();
    }
}
```

Task 12:

What are access modifiers in java … give detailed explanation?

Access Modifiers is keywords used to control the accessibility of classes, variables, methods, and constructors. They help to implement data hiding and encapsulation.

There are four types of access modifiers:

1. Public – Accessible from anywhere
2. Protected - accessible in same packages and subclasses
3. default (no modifier) – No keywords used, Accessible only in same packages
4. Private – Accessible only inside the same class.

Task 13:

Write a program to accept cust names and display them using for each loop

```
import java.util.Scanner;

public class cust_name{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of customers: ");
        int n = sc.nextInt();
        sc.nextLine(); // Consume newline

        String[] customers = new String[n];

        // Accept customer names
        for (int i = 0; i < n; i++) {
            System.out.print("Enter customer name " + (i + 1) + ": ");
            customers[i] = sc.nextLine();
        }

        // Display customer names using for-each loop
        System.out.println("\nCustomer Names:");
        for (String name : customers) {
```

```
            System.out.println(name);
        }

        sc.close();
    }
}
```

Task 14:

Write a program to show the difference between break and continue.

Use for loop to show the difference

```
public class BREAK_CONTINUE {
    public static void main(String[] args) {

        for (int i = 1; i <= 10; i++) {
            if (i == 5) {
                break;
            }
            System.out.println(i);
        }

        System.out.println("Loop terminated.");


        for (int i = 1; i <= 10; i++) {
            if (i == 5) {
                continue;
            }
            System.out.println(i);


        }
    }
}
```

Task 15

```
class Employee {
    int empId;
    String empName;
    double salary;

    Employee(int empId, String empName, double salary) {
        this.empId = empId;
        this.empName = empName;
        this.salary = salary;

    }
}

class Manager extends Employee {
    String department;
```

```
    Manager(int empId, String empName, double salary, String department) {
        super(empId, empName, salary);
        this.department = department;
    }

    void display() {
        System.out.println("Employee ID: " + super.empId);
        System.out.println("Employee Name: " + super.empName);
        System.out.println("Salary: " + super.salary);
        System.out.println("Department: " + department);
    }
}

public class task15 {
    public static void main(String[] args) {
        Manager m = new Manager(101, "Monika", 50000, "HR");
        m.display();
    }
}
```

Task 16

Public private protected

```
class Employee1 {
    public String name = "Monika";
    private double salary = 50000;
    protected int age = 25;

    public void displaySalary() {
        System.out.println("Salary: " + salary);
    }
}


public class public_private {
    public static void main(String[] args) {
        Employee1 e = new Employee1();

        System.out.println("Name: " + e.name);
        System.out.println("Age: " + e.age);
        // System.out.println("Salary: " + e.salary);
        e.displaySalary();


    }
}
```